Topic 14: Switch Node & Workflow Routing — Implementation Document

1. Implementation Overview

The workflow uses n8n visual nodes to implement multi-branch routing based on user roles.

2. Workflow Nodes

1. Manual Trigger
2. Input User Profiles (Edit Fields)
3. Split Out
4. Route Users by Role (Switch)
5. Set Admin Status (Edit Fields)
6. Set Editor Status (Edit Fields)
7. Set Viewer Status (Edit Fields)
8. Set Guest Status (Edit Fields)

3. Node Configuration

3.1 Manual Trigger

Add a Manual Trigger node to start workflow execution.

3.2 Input User Profiles

Configure an Edit Fields node in JSON mode.

Input:

```
None
{
  "profiles": [
    {
      "name": "Alex",
      "role": "admin"
    },
```

```
    {
      "name": "Sarah",
      "role": "editor"
    },
    {
      "name": "John",
      "role": "viewer"
    },
    {
      "name": "David",
      "role": "guest"
    }
  ]
}
```

3.3 Split Out

Configure:

- Fields to Split Out: `profiles`
- Include: No Other Fields

Expected result: Four separate JSON items.

3.4 Switch Node

Node name: Route Users by Role

Configuration:

- Mode: Rules
- Value 1: `{{ $json.role }}`
- Operator: is equal to

Rules:

Rule	Value
Rule 0	admin

Rule 1	editor
Rule 2	viewer

Fallback Output: Extra Output.

3.5 Set Admin Status

- Field Name: `isAdmin`
- Type: Boolean
- Value: `true`

Connect to Switch Output 0.

3.6 Set Editor Status

- Field Name: `isEditor`
- Type: Boolean
- Value: `true`

Connect to Switch Output 1.

3.7 Set Viewer Status

- Field Name: `isViewer`
- Type: Boolean
- Value: `true`

Connect to Switch Output 2.

3.8 Set Guest Status

- Field Name: `isGuest`
- Type: Boolean
- Value: `true`

Connect to Switch Fallback Output.

4. Execution Procedure

1. Save the workflow.
2. Execute the Manual Trigger.

3. Verify the input profile data.
4. Execute the Split Out node.
5. Confirm four individual items.
6. Execute the Switch node.
7. Verify all four output branches.
8. Execute each downstream Edit Fields node.
9. Verify the Boolean output values.

5. Expected Execution Results

Profile	Role	Branch
Alex	admin	Output 0
Sarah	editor	Output 1
John	viewer	Output 2
David	guest	Fallback

6. Implementation Conclusion

The workflow successfully implements multi-way routing using the n8n Switch node. The default fallback branch handles unmatched roles without requiring nested IF nodes or custom JavaScript.